



Enforcement of the EU greenhouse gas emissions trading scheme

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1. INTRODUCTION

One of the remarkable developments in environmental law is the evolution of emissions trading from a theoretical concept towards a main tool of actual climate change policies. The inclusion into the Kyoto Protocol of inter-state emissions trading (ISET) and project-based crediting of emissions reductions (joint implementation (JI) and clean development mechanisms (CDM)) is, indeed, an important milestone of international environmental law. In addition, the European Union (EU) has established the largest domestic emissions trading scheme worldwide, which actually started on 1 January 2005.² It will be interesting to see whether this application of emissions trading will have the expected positive results in practice. Outcome assessments are needed in order to learn about the practical effects of the instrument in different respects, and to facilitate discussions about improving the present scheme.

One core obligation of an emissions trading scheme (ETS) is that each covered installation needs to surrender an amount of emissions rights at least equal to the emissions during a certain well-defined period. In this respect, close attention needs to be paid to the compliance behaviour of firms. It is obvious that compliance with the rules of the emissions trading scheme is a precondition for letting that market work. When some firms would act in breach with the rules, the environmental effectiveness would become less, and the trust within the market would become at risk. Enforcement issues of emissions trading deserve especially close attention as it can be supposed that it might be financially attractive for firms to behave illegally. Furthermore, the fact that enforcement activities towards industries are to be executed by the Member States leads to considerations to harmonize to a certain extent the enforcement provisions at the EU level. Different enforcement strategies (strong and weak strategies) among the Member States could affect the level playing field between industries, and could lead to a distortion of the market for greenhouse gas (GHG) allowances.

In this chapter, it will be discussed how the EU GHG emissions trading programme deals with the fact that compliance by the covered industry must be

ensured. In its Green Paper on GHG emissions trading within the EU, the European Commission (the Commission) already recognized the need for a robust enforcement regime:

The purpose of strict compliance provisions and enforcement is to enhance confidence in the trading system, make it work in an efficient way in accordance with the rules of the internal market and at the same time increase the likelihood of achieving the desired environmental result.³

The Green Paper stressed the need for adequate monitoring, tracking of the allowances, and the need of reporting, verification and control. In addition, strong penalties should be established; they should be foreseeable and must significantly exceed the cost of compliance, according to the Commission. Subsequently, several provisions for monitoring, verification and enforcement are included in the Greenhouse Gas Emissions Trading Directive (GHG ET Directive). Those provisions, which will be discussed below, have harmonized the enforcement approaches by Member States to a large extent.

Within this chapter, section two will first outline some general preliminary remarks. Section 3 will examine which obligations are established for the covered industry under the present greenhouse gas emissions trading scheme (GHG ETS). Section four will then discuss the enforcement package as included within the EU ET Directive, and, finally, section five will end with an outlook forecast.

Besides focusing on (enforcing) compliance by industry, the level of compliance by the Member States in fulfilling their duty to implement the ET Directive is, of course, essential for letting the market work. This chapter will, however, focus on the enforcement issues with regard to industry. Within chapter three issues of Member State liability when failing to implement sufficiently the ET Directive have already been discussed.

2. GENERAL REMARKS

2.1 The Need for a Strong Enforcement Package

The rationale for introducing emissions trading is that a price-driven decision-making by potential polluters will be enhanced, facilitating the covered industry (or other sectors of society to be covered) to choose between reducing emissions at a certain cost, or getting emission rights from the market, or, depending on the specific design of a scheme, from the government. Of course, within the first allocation of the allowances, a great deal or all of the available rights may be given for free to the industry (this is often called ‘grandfathering’). This is in fact what happened during the first allocation round within the EU ETS. Nev-

ertheless, also in this case a market incentive will be included as firms are in the position to choose between (at least): (1) reducing their emissions and selling their allowances; (2) using their allowances in order to cover their emissions; or, (3) increasing emissions and buying extra allowances on the market. The market incentive will differ depending on the specific design of the permit-market. However, in any ETS a certain market incentive is automatically included, and is thus in fact to be seen as a basic feature of emissions trading. The price of the allowances not only creates an incentive to reach environmental protection at low costs, but serves at the same time as an incentive for firms to seek the loop-holes within the rules. It is even possible that some firms would deliberately choose to be in non-compliance, when that would be profitable and when, for instance, the chance to be detected is presumed as being low. Hence, the main and most attractive feature of market-based regulation – establishing price-driven decision-making by industry with regard to their environmental behaviour – is, in fact, also a threat for the level of compliance. It is clear that adequate enforcement provisions must be established and applied in order to back up the permit market. Those provisions need already to be part of the law establishing the permit scheme. One can particularly think of the following provisions:

- *Clear and sound legislation* – it is important that the rules are crystal-clear and hence don't lead to different points of view whether an activity must be qualified as in breach with the law.
- *Monitoring, reporting and control provisions* – in order to verify how much allowances need to be surrendered.
- *Adequate sanctions* – the preventive function of the sanctions is of course important. However, the possibility to restore through sanctioning the over-emissions in case of non-compliance also deserves close attention, in order to ensure the environmental effectiveness of the scheme.

In this contribution, the notion 'enforcement' refers to this whole package of provisions.

2.2 The Need for Specific Enforcement Provisions for Emissions Trading

In general, the monitoring and control issues in the case of emissions trading differ from the classical command and control regulations. The classical way of enforcement in the case of command and control depends on *static* obligations for firms, like binding limit values for their emissions, or the obligatory application of specific techniques. In fact, within the classical command and control permits the obligations for covered firms do not change unless the

permit is amended (as in the case of the Integrated Pollution and Prevention Control (IPPC)-permit). With emissions trading however, the legal obligation for covered firms is a *dynamic* one: the amount of allowances that need to be surrendered fluctuates depending on the exact amount of emitted gases. This means that the control and enforcement provisions must be adapted to this dynamic obligation. The measurement of the fluctuating emissions is crucial in order to determine the amount of allowances that need to be surrendered by the operator of the installation. It in fact means that a continuous control provision for measuring and/or accounting the emissions is inevitable, ie as has been installed within the acid rain allowance trading programme that runs in the USA.

2.3 Towards Multi-pollutant Emissions Trading?

The present GHG ETS actually only covers CO₂ emissions. This approach runs counter to the idea that environmental law should have a comprehensive approach. The challenge for the future is to examine the possibilities of multi-pollutant emissions trading systems, including the greenhouse gases and even also other air pollutants. One of the main questions is, of course, whether a multi-pollutant trading scheme can indeed be controlled effectively. CO₂ emissions seem rather easy to monitor – so here big problems would hopefully not occur with the design of monitoring requirements within the GHG permit. Expansion of the ETS can be considered in order to enhance cost-effectiveness in climate change policies.

Nevertheless, the monitoring then may become more complicated, as then pollutant equivalents must be taken into account. Hence, it needs to be agreed upon whether emissions trading will still be a preferable regulatory instrument for a multi-pollutant approach. This approach is in fact already included within the Kyoto Protocol. It is, however, not yet clear whether the monitoring issue of multi-pollutant trading can indeed be solved for the national or regional schemes for GHG emissions trading.

3. LEGAL OBLIGATIONS FOR INDUSTRY WITHIN THE GHG ETS

Before the necessary enforcement provisions can be discussed, we first need to identify what the main possible offences by industry can be within the EU GHG ETS. It appears that the GHG emissions permit, as being established by the directive, forms the crucial legal tool for specifying the obligations of industry. Actually two core obligations for industry can be distinguished:

1. Greenhouse gases may only be emitted by the covered industry when a greenhouse gas permit has been issued.
 2. To comply with the greenhouse gas permit conditions.
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1. Greenhouse gases may only be emitted by the covered industry when a GHG permit has been issued.

Besides the transferable allowances, to be allocated by the national governments and which may be sold on the emissions trading market, a so-called 'greenhouse gas permit' is included within the GHG ETS Directive. Greenhouse gas emissions (during the first allocation phase running from 2005 till 2008 this concerns only CO₂) may only be emitted by an installation when the operator holds a GHG emissions permit.⁴ The permit is meant as a legal document in which the specific GHG activities of the installation(s) are described, and in which main obligations for the operators are included. According to the Directive, the permit may only be issued by the competent authority if it is satisfied that the operator is capable of monitoring and reporting emissions.⁵

2. To comply with the GHG permit conditions

The GHG permit is, in fact, an important tool for ensuring compliance with the scheme. Within the permit, monitoring and reporting requirements need to be included.⁶ Moreover, the permit must contain the obligation for the operator to surrender allowances equal to the total emissions of the installation in each calendar year, within 4 months following the end of that year.⁷

The operator is obliged to inform the competent authority of any changes planned in the nature or functioning, or an extension, of the installation that might require updating of the GHG emissions permit.⁸ Of course, this rule needs also to be monitored and enforced by the competent authority.

The specific requirement mentioned above, that the permitting authority must be satisfied that the operator is capable of monitoring and reporting requirements, will not always be easy to execute in practice. It can be assumed that bad behaviour in the past – especially non-compliance with monitoring and reporting obligations – could be a reason for not having enough trust in the capabilities of the operator. However, the denial of a permit request based on the ground that the operator probably would not be able to comply with the monitoring and reporting emissions needs a sound argument. Such a denial would bring a firm into the unfortunate position that it would not be allowed to emit greenhouse gases anymore, which could mean in fact that the whole unit would no longer be able to operate. Such a high consequence needs a very strong justification.

4. THE ENFORCEMENT PACKAGE OF THE EU GHG ETS

4.1 Monitoring, Reporting and Verification

4.1.1 Introduction: a harmonized approach

The GHG ET Directive provides for an extensive framework for the monitoring, reporting and – to a lesser extent – the verification provisions to be implemented by the Member States. Nevertheless, as will be discussed below, some discretion is given to the Member States and their competent authorities in deciding on the specific monitoring, reporting and verification duties. Such discretion is necessary in order to make the system pragmatic and workable, and, moreover, to respect the autonomy and the specific legal cultures of the Member States. On the other hand, it could be the case that this discretion – when not implemented in a sound way – would be a risk for the environmental effectiveness of the European ETS.

Article 14 of the Directive obliges the Commission to adopt ‘guidelines’ for the monitoring and reporting of emissions by the industry. These guidelines need to be based on the principles for monitoring and reporting as set out in Annex IV of the Directive. The Member States are obliged to ensure that emissions are monitored in accordance with these guidelines, and must ensure that each operator of an installation reports the emissions from that installation during each calendar year to the competent authority after the end of that year, in accordance with the guidelines. Considering these provisions within the Directive, it can be argued that the guidelines of the Commission have a rather strong legal meaning.⁹ According to the Commission, Member States are, indeed, not allowed to give derogations to the application of the guidelines unless such derogations are explicitly provided for *in* the guidelines.¹⁰ However, the guidelines as such give some discretion for deciding on the specific duties for industries.

4.1.2 Balancing between accuracy and costs

According to the Directive, the Commission had to adopt the guidelines by 30 September 2003, which, remarkably, is even before the Directive entered into force.¹¹ The actual, the guidelines were adopted on 29 January 2004.¹² The monitoring and reporting guidelines set out general guidelines (in Annex I) and activity-specific guidelines (in several annexes). The principles (included in Annex IV of the Directive) state that carbon dioxide emissions shall be monitored either by calculation or on the basis of measurement. For the calculation of the emissions a formula is prescribed (Activity data × Emission factor × Oxidation factor).

Of course, the specification of the exact monitoring methodology to be used by the operator is an important element of the enforcement package. The guide-

lines stress not only the accuracy and completeness of the data, but also the cost-effectiveness of a monitoring methodology: monitoring and reporting of emissions should be done with the highest achievable accuracy, unless this is technically not feasible or will lead to unreasonable high costs.¹³ The monitoring and reporting guidelines of the Commission thus try to balance between the problem of technological uncertainties in monitoring emissions and the need for accurate reporting. In the specific annexes, several tiers of approaches are presented, representing different levels of accuracy. An operator has to choose the tier with the highest level of accuracy. The guidelines point out that only if it is shown to the satisfaction of the competent authority that the highest tier is technically not feasible or will lead to unreasonably high costs, the next lower tier may be used, subject to approval by the competent authority.¹⁴ According to the guidelines, lower tiers may be applied for minor sources, which would also need the approval of the competent authority.¹⁵

The assessment of whether specific costs of a monitoring methodology would be unreasonable can result in a conflict between operators and administrative authorities. For a good understanding of the environmental integrity of the scheme, and also of the (comparable) costs of this directive for industry, it will be interesting to assess what the content of the decisions (with respect to the monitoring methodologies) are within different Member States, and whether those monitoring decisions are rather similar, especially when it would concern comparable industrial installations.

The exact degree of harmonization by these guidelines of monitoring and reporting within the EU depends on the wording and details thereof – and it can be concluded that the guidelines do indeed leave some discretion to the Member States, especially where it concerns the specification of the exact monitoring methodologies.

4.1.3 The permit

The monitoring and reporting guidelines point out that the competent authority shall approve a detailed description of the monitoring methodology prepared by the operator ‘before the start of the reporting period, and again after any change to the monitoring methodology applied to an installation’.¹⁶ The directive prescribes, in art. 6(2)(c), that the GHG emissions permit shall contain the monitoring requirements, specifying the monitoring methodology and frequency. This means that, after the initial issuance of the GHG permit, changes towards the monitoring methodologies need to be approved by the permitting authority, and to be included within the updated GHG permit.¹⁷ However, the guidelines suggest that, ‘where consistent with the Directive’ the monitoring methodology will be specified in general binding rules. According to art. 6(2)(b) of the Directive, the GHG permit must be seen as the central place for these requirements. Also, according to art. 7 of the Directive, the operator shall inform

the competent authority of ‘any’ changes planned in the nature or functioning of the installation, which may require updating of the GHG permit. An update of the monitoring requirements could be qualified as being such a change that would require an update of the GHG permit. Hence, it is questionable whether and how Member States could make use of the general rules, as suggested by the guidelines.

4.1.4 Reporting

In addition to the monitoring requirements, the permit must contain ‘reporting requirements’, according to art. 6(2)(d) of the Directive. The guidelines of the Commission also refer to reporting. The Member States have to ensure that each operator of an installation reports the emissions from that installation during each calendar year to the competent authority after the end of that year in accordance with these guidelines.¹⁸ According to Annex IV of the Directive, on which the guidelines must be based, the reporting requirements need to refer to data identifying the installation, specific data referring to the calculation of emissions and the measurement of emissions, and some specific data for emissions from combustion. According to Annex IV, in the case of reporting data, with respect to the calculation or measurement of emissions, any ‘uncertainty’ has to be pointed out, and specific for measurement, ‘information on the reliability of measurement methods’ should be included. It can be expected that these kinds of points of attention can lead to different opinions between operators and the authorities concerned, and thus will be a (probably to a certain extent unavoidable) source of legal conflict.

4.1.5 Uncertainty about the data, and misstatements

Following the guidelines there is some room for uncertainty about the data, or even misstatements. This can be illustrated by the following terms as included in the guidelines:

‘As a general rule the guidelines specify a “permissible uncertainty”, which is expressed as “the 95% confidence interval around the measured value”.’¹⁹

The monitoring and reporting guidelines introduce the term ‘materiality’: ‘materiality means the professional judgement of the verifier as to whether an individual or aggregation of omissions, misrepresentations or errors that affects the information reported for an installation will reasonably influence the intended users decisions. As a broad guide, a verifier will tend to class a misstatement in the total emissions figure as being material if it leads to aggregate omissions, misrepresentations or errors in the total emissions figure being greater than 5 percent’. The guidelines express that the verifier has to assess the materiality of the reported facts. When the verifier concludes that the emissions

report does not contain 'any material misstatement', (and it can be assumed that this assessment includes the margin as specified in the definition of materiality), the report is verified as satisfactory. The operator can then submit the report to the competent authority.

Of course, the question arises as to whether operators would try to gain as much profit from the room for interpretations as been given by the Directive and the guidelines. Principally, one would prefer that not any error or mistake would be accepted, so that no room for misstatements would be included. It is to be recalled that art. 6(2)(e) of the Directive states that there is an obligation to surrender allowances *equal* to the total emissions of the installation in each year, as verified. Annex V states that a high degree of certainty needs to be achieved. Full certainty in every case is probably not feasible in practice. However, the assessment of 'materiality' can obviously lead to legal disputes, and accepting a rather broad margin of errors would be in conflict with the provisions of the Directive.

4.1.6 Verification

The annual reports have to be verified in accordance with criteria set out in Annex V. The Directive does not oblige the Commission to set guidelines for verification, but the published guidelines do, in fact, refer to a certain extent to this.

The verification has to be accomplished by 31 March each year for emissions during the preceding year. A strict sanction has to be applied automatically by the Member States when this deadline is passed: the Member States must ensure that such an operator cannot make further transfers of allowances until a report from that operator has been verified as satisfactory.²⁰ Here it can be questioned what the term 'transfer' exactly means, as it has not been defined in the Directive. If it only means the selling of allowances, then the ability to buy allowances in order to cover ex post its emissions during the preceding year would still be available to the operator. According to art. 16(3) of the Directive, any operator must surrender sufficient allowances by 30 April of each year to government. It seems reasonable to interpret the term 'transfer' in this respect, so it would only mean that an operator is only prohibited to sell any allowance until he has got a satisfactorily verification.

The Directive clearly sees the possibility that non-governmental verifiers will act. Member States may choose whether or not to attribute the verification task to administrative authorities, or to independent private verifiers. In any case, the verifier must carry out its activities in a 'sound and objective professional matter'. A practical question (with possibly legal consequences) is whether there will be enough (qualified) verifiers, at reasonable prices, who are able to verify in time. Member States (and industry and advisors) should aim at organizing adequate verification possibilities. It is not necessarily the case that the costs of

verification must be carried by the operators (s this is not indicated in the Directive) but it can be assumed that, to a large extent, the operators will have to pay for the verification. This can depend on the choice of the Member States whether to allow private verifiers, or to execute this task themselves (although it even then is not excluded that a fee will be charged).

Since the reports only have to be presented *after* a calendar year, it can become problematic to control *afterwards* whether the presented report is true. Hence, the verification process seems to be only adequate if it also concerns the credibility of the monitoring *during* the preceding year. Only *ex post* verification – on the basis of a report – would not be sufficient. A more or less continuous verification – by, for instance, random checks – during the whole period of monitoring is necessary, although probably rather costly. It is clear that the verification process must be accompanied by site inspections in order to investigate whether the presented data or facts are the same as in practice. According to Annex V, the verifier shall indeed be given access to all sites and information in relation to the subject of the verification. In addition, the following criteria have been provided for:

‘The verification of the information submitted shall, where appropriate, be carried out on the site of the installation. The verifier shall use spot-check to determine the reliability of the reported data and information.’²¹

This provision is an important one, and according to Annex V(11), the verifier must make clear in its verification report how the verification has been executed, including information on the executed site inspections.

One question posed is whether a verifier will indeed be a credible one? Annex V states that the verifier shall be independent of the operator. It can be assumed that some provisions are necessary in order to make the verifiers as credible as possible, like the accreditation of private verifiers by government.

The case could arise that a verifier has stated that the report is satisfactory, but that the government would doubt whether the verifier is – in that specific case – really independent. Doubt can also exist about whether the verification process has been executed in a professional way. It can not be excluded that a situation might occur where a government has evidence that the verifier made an error in its assessment. Hence, in order to comply – as a Member State – with the obligations laid down in the Emissions Trading Directive, additional competences for administrative authorities are required to overrule the findings of the verifier. This also means that administrative authorities need competences for in-site inspections. Of course, it is extremely important that the verification tasks will be carried out sufficiently by the verifiers, but – as a second and additional inspection provision – the government is not dismissed from the duty to control whether the rules of the emissions trading framework will be complied with.

The relationship between the operator and a verifier also deserves attention. For instance, what is the legal position of a firm that has presented in time a report to a verifier, but where subsequently the verifier appears not to be able – for instance, due to internal bad organization – to verify in time? In such a case, the consequences are heavy for the firm, as it will mean an offence of the rule laid down in art. 6(2)(e) of the Directive specifying the obligation to surrender allowances equal to the total of emissions of the installation in each calendar year, as verified, within 4 months following the end of that year. In addition, the firm will not be allowed to take part of emissions trading (meaning in particular the selling of allowances). It can be recommended to operators, contracting a verifier, to refer to this possible situation in the contract with the verifier, and to claim a financial compensation for the damage occurred. The Directive does not give any discretion to the Member States in suspending the requirement that by 31 March of each year the report must have been verified as satisfactory. One could suppose that in cases of force majeure a power to suspend would be reasonable.

In addition, it also needs to be taken into account that the integrity of the civil servants and administrators is not naturally guaranteed. Unfortunately, we have learned that examples of fraud exist within administrative institutions as well. Thus, as it can be financially attractive to help industry with cheating or falsifying data, the integrity of the administration also needs attention.

4.1.7 Conclusion

In summary, it can be concluded that the GHG ET Directive has harmonized the monitoring and reporting provisions to a rather large extent, and has also given some direction towards the verification provisions. A careful approach by the Member States is needed in implementing the monitoring, reporting and verification obligations of the Directive – thereby filling in the room for discretion left by the Directive and the guidelines. It is clear that important administrative action needs to be taken, such as the issuance of the GHG permit (including an assessment of the capability of the operator to monitor and to report), and the specification of monitoring and reporting requirements – thereby following the guidelines of the Commission. In addition, a sound credible verification process must be facilitated. Above all, the monitoring requirements for industry must be controlled effectively.

The issuance and update of the GHG permit is a crucial step, because in this document essential provisions must be included. Legal disputes might arise between the operators and the administration with respect to several decisions and assessments that need to be taken within the monitoring, reporting and verification provisions. It can, for instance, certainly not be excluded that with choosing the methodology dissenting opinions may rise between the operator and the competent authority, and that subsequently legal disputes may thus occur in the permitting process with regard to the monitoring methodology. Also, when

the monitoring methodology has been specified, it is to be expected that operators won't always follow this legal obligation. For instance, false data may be used, or errors may be made in calculations or measurements.

Disputes might even arise between the operators and the private verifiers, and diverging assessments between a private verifier – and the ultimate responsible administrative authority are also not to be excluded. The authorities remain ultimately responsible in checking whether the monitoring activities are sound. Unannounced random checks by authorities will be an inevitable part of the enforcement package.

Ultimately, much will depend on the practical control strategies developed and executed by the Member States. It would not be surprising to see Member States would completing the given framework for monitoring, reporting and verification with different levels of ambition. A basic question in this respect is whether the Member States will provide for enough financial capacity for executing the necessary control tasks, such as, for instance, making regular on-site inspections by competent civil officers.

4.2 Sanctions

4.2.1 Introduction

Most of the Member States have practically no experience with emissions trading, and thus no experience with establishing sanctions for breaches with the legislative obligations that are part of an ETS. The Directive seems to support the legislators of the Member States to a certain extent, as it specifies several specific sanctions in art. 16. Besides the well-known and common prescription that the penalties provided for must be effective, proportionate and dissuasive, some particular sanctions are prescribed which need to be implemented by the Member States.

4.2.2 Naming and shaming

The first penalty mentioned in art. 16 is a remarkable one:

Member States shall ensure publication of the names of operators who are in breach of requirements to surrender sufficient allowances under Article 12(3).²²

This provision is, in fact, not strictly needed for enforcing an ETS. More remarkably, this provision marks a new enforcement approach at least in the field of European environmental legislation. The duty to publish the names of offenders can be called a 'naming and shaming' approach, and has not yet been included in European secondary environmental legislation.

With the Emissions Trading Directive, the Member States will thus be obliged to 'name and shame' offenders. About naming and shaming, especially as a

criminal sanction, an intense academic debate has already been held, and there is discussion about the moral aspects and – more practically seen – the effectiveness and usefulness of the approach.²³ It has even been argued that name and shame sanctions would be an unacceptable style of governance.²⁴ Introducing this apparently questionable enforcement approach within European environmental law would require a careful discussion, which does not seem to have taken place. The explanatory memorandum by the proposal of the Commission does at least not justify why this new approach has been chosen.

In addition, the implementation of the ‘naming and shaming’ provision embodies interpretative questions. For instance, at what time is it reasonable to publish? Immediately at 1 May each year, when according to art. 12(3) of the Directive every operator had to surrender a number of allowances equal to the total emissions? As the ‘naming and shaming’ approach is presented as a penalty, would not it be necessary to consider the specific circumstances under which the operator was not able to surrender the allowances, such as the question of whether or not it is the operator or probably the verifier (or even the government) who didn’t act in time? Also, would not it be preferable to publish the name after a judge has stated that indeed the operator acted culpable or deliberately in breach of art. 12(3)?

However, the advantages of naming and shaming must be taken into account as well. Within the US acid rain allowance trading programme, information technology has been used for making the information transparent. It is suggested that making the information derived from the continuous emissions monitoring system online available to the public would increase compliance.²⁵ Such positive effects should, of course, be taken into account when assessing the necessity and acceptability of a naming and shaming approach. However, the instrument has now already been prescribed to the Member States, unfortunately without an extensive discussion on the rationale for introducing this, in fact, challenging instrument.

4.2.2 Financial penalty

The second penalty the Member States must implement is a fixed and automatic penalty, which is called the ‘excess emissions penalty’.²⁶

The penalty is ‘fixed’, since the Directive prescribes the amount to be paid by the offender. For the first phase 2005–07, the excess emissions penalty will be €40 for each ton of carbon dioxide equivalent emitted for which no allowances have been surrendered by 30 April of a calendar year. From 2008 onwards, the penalty is fixed at €100 per ton of carbon dioxide equivalent.

The proposal for the directive from the Commission included another approach: the excess emissions penalty would be €100 ‘or either twice the average market price between 1 January and 1 March of that year for allowances valid for emissions during the preceding year’. Within this idea, a connection with

the prevailing market price would be provided, which is a more flexible approach.

Of course, the core question is whether or not the penalty would be severe enough to deter possible offenders. The experience with the acid rain emissions trading programme – which has a fixed and indexed penalty price – has shown that the established penalty is far more than twice the market price. In fact, within that programme the penalty is 10 and even (approximately) 20 times higher than the average market price of the allowances.²⁷ The level of the penalty within the European ETS doesn't *seem* to be very high but before concluding on this one should know what the market price of the allowances will be, which is probably hard to predict.²⁸ On the other hand, it has to be considered that, in other literature, it has been argued that a high penalty is not necessarily always effective.²⁹ Additional sanctions, such as imprisonment, would then be needed.

Even more remarkable is the fact that the Directive prescribes that the penalty in any case should be imposed. The financial penalty is thus 'automatic', because art. 16(3) of the Directive prescribes that this excess emissions penalty has to be paid by 'any operator who does not surrender sufficient allowances by 30 April of each year'. It would be more proportionate when the circumstances that lead to the breach with the rule would be taken into consideration by the enforcement authority. The Directive, however, does not specify what leeway Member States would have in applying the financial sanction to operators. On the other hand, discretion in defining the level of the penalty embodies the danger of sparing the industry. From a legal point of view a careful case-by-case consideration of whether the penalty indeed must be imposed would be preferable instead of having an automatic system of sanctioning. It can be expected that through jurisprudence the order to impose the penalty in an automatic way, which vests in fact a risk liability to be followed by a criminal sanction, will mitigate this provision.

The Directive also does not indicate whether an administrative authority, or a judge, must impose that penalty. It can be assumed that the most practical way will be to give an administrative authority the competence to impose the penalty, whose decision then can be appealed in court.

4.2.3 Compensation of excess allowances

In addition to the excess emissions penalty, the operator is still obliged to surrender an amount of allowances equal to the excess emissions.³⁰ This enforcement provision is needed in order to uphold the environmental effectiveness of the ETS.

4.2.4 Prohibition to sell allowances

Another 'penalty' has already been discussed above: in case of not having a satisfactorily verified report, further transfers (meaning selling) of allowances

are not allowed. It is assumed that operators would be still in the position to buy allowances, in order to fulfil the requirement to surrender an amount of allowances equal to the excess emissions. The prohibition to sell excess allowances will, of course, not be felt as a sanction in cases where there are not enough allowances are held compared to the actual emissions.

4.2.5 Extra sanctions, not prescribed by the Directive

When an operator would have breached rules on monitoring and reporting obligations, the situation might occur where no or wrong data are available that would state or indicate the amount of emissions during a calendar year. In this situation the basic obligation of emissions trading (which is the rule that allowances have to be surrendered that are equal to the emissions) can not be applied, because the emissions data are missing. The excess emissions penalty as provided for in the Directive would then of course not be applicable, since with a lack of data it is not known how many allowances must be surrendered and penalized. The Directive does not include a provision for this specific but not unimaginable situation. The Member States thus need to develop specific enforcement competences for this situation. What can be considered by the national legislator that is to make the authorities competent to estimate the amount of emissions in a calendar year concerned, and that subsequently a penalty on the basis of that estimation will be applied. Of course, it can be rather difficult for the enforcing authority to assess the just level of the penalty, and the assessment can be brought to court, but this approach is at least an attempt to deal with this offence and should be part of the Member States legislation, until a better approach can be found.

It is clear that additional enforcement provisions, like the one discussed above, must be considered by the Member States. For instance, sanctions for the situation where an operator fails to report in time or presents a false report, penalties for acting without an (amended) permit (which could ultimately be the closing of the installation, when proportional), or penalties for breaches of the monitoring requirements should be established.

The question also arises as to whether additional prison sentences must be installed by the Member States. In addition, a provision concerning the removal of illegal gains could be part of an enforcement strategy. The duty to surrender allowances for the excess emissions already constitutes a part of such a strategy.

5. CONCLUSION AND OUTLOOK

In order to reach the goals of European climate change policy, and to comply with the Kyoto commitments and burden-sharing commitments for the EU and

its Member States, it will be of crucial relevance that the GHG ETS will function effectively. The remarkably fast adoption of an EU-wide emissions trading directive is clearly not enough; a successful outcome (and other climate change measures) is needed in order to show – in the global arena – that Europe indeed takes its responsibility seriously.³¹ First points of attention in this respect are: (1) a sound implementation of the ET Directive by the Member States, including national allocation plans that seriously take into account the need to decrease the CO₂ emissions; and (2) the issuance of GHG permits with adequate control provisions. Then, ideally, 100% compliance by the covered industry with the core obligations of the ETS is needed in order to ensure the environmental effectiveness of the scheme.

It is up to Europe now to show that, in our complicated and decentralized EU-law system, a high rate of compliance will occur, and – and that when this would be lacking – that non-compliance in most cases will be detected, restored and penalized. This is probably not too difficult in the first allocation phase of the scheme where the emissions reductions asked from the covered industry are not severe. The issue of compliance will be a weightier question in the following phases, where more serious emissions reductions would be included in the national allocation plans.

The first phase of the EU ETS can thus be seen as a learning phase, providing the possibility to gain experience with this instrument. According to art. 21 of the ET Directive, the Member States are obliged to report every year on the application of the Directive. In the report particular attention must be paid to:

- The application of the monitoring and reporting guidelines;
- The verification and issues relating to compliance with the Directive.

This means that, every year, important information will (or should) become available on the compliance issues with the Directive. The first reports need to be sent in by 30 June 2005, and these reports should already give information on compliance with the duty to have a GHG permit when emitting CO₂, and on other parts of the enforcement package for emissions trading.³²

Subsequently, on 30 June 2006, the Commission is expected to submit a report to the European Parliament and the Council on the application of this Directive.³³ The report must be accompanied by proposals as appropriate, obviously in order to improve the system. In the list of topics that need to be considered by the Commission in this review process, little has been included on enforcement issues. The Commission is only obliged to consider the level of excess penalties, taking into account, *inter alia*, inflation. Remarkably, nothing has been said on topics such as monitoring provisions, reporting duties, verification, and alternative approaches to sanctioning, (*ie* connecting the penalty to the prevailing market price). Nevertheless, on the basis of the yearly reports, and

independent assessments by the scientific community or other organizations (and probably on the basis of case-law), it will become clear whether current provisions ought to be revised.

The enforcement package of the EU ETS is widely harmonized, but it is clear that the real actions need to be taken primarily within the Member States. Member States have to fulfil an important role in ensuring the environmental integrity of the ETS. They need to implement the ET Directive and to control and to enforce industry where necessary. The active role of Member States – for instance, with setting up enforcement strategies – is not only important for the environmental effectiveness of the scheme, but also for ensuring as much as possible the competition concerns of the covered industry, often qualified as the concern for a ‘level playing field’.

However, one must not overlook that within the transboundary, EU-wide emissions trading market, the Member States are in charge and control their own industry. A close eye on the way in which Member States execute this task is strongly recommended. In fact, Member States will also have an economic incentive not to be too severe towards their own industry. The most important enforcement task is, therefore, probably the task of the Commission controlling the enforcement duties of the Member States, and to start an infringement procedure when necessary.

A lot of the provisions belonging to the enforcement package are harmonized by the ET Directive or the binding guidelines, such as the duty to have the GHG permit, the monitoring and reporting provisions, and the sanctions. However, this does not guarantee that an intensive enforcement strategy will be followed by the Member States. Indeed, a further harmonization or even centralization of the more practical enforcement aspects of the ETS would cause an intense debate, and does not seem in line with the principle of subsidiarity. Nevertheless, it can be argued that EU-wide assessments should be made of the content and intensity of the enforcement strategies followed by the Member States in practice. That, at least, gives an impression on how tough the Member States do control their own industries, and how many on-site inspections, for instance, take place. The reports which need to be sent to the Commission by the Member States (according to art. 25 of the Directive) are expected to provide important information with regard to this.³⁴ It can even be considered to extend the questionnaire with specific questions on the practical aspects of enforcement, in order to gain a deep insight into the compliance behaviour by industry and the intensity of the enforcement strategies of the Member States with respect to emissions trading.

An ultimate, though for the moment not very realistic, provision would be that the inspection tasks would not be established only within the Member States, but that the Commission would have additional inspection competences.³⁵ Such a far-reaching option of course is an ultimate option, only to be taken

when the Member States would appear not to be capable of ensuring the enforcement integrity of the GHG ETS.

NOTES

1. Associate Professor of transboundary environmental law, Maastricht University, Faculty of Law, Metro Institute (e-mail: Marjan.peeters@pubr.unimaas.nl).
2. Directive 2003/87/EC of the European Parliament and of the Council of 13 October 2003, establishing a scheme for greenhouse gas emission allowance trading within the Community and amending Council Directive 96/61/EC, OJ EU L 275/32, 25 October 2003. This Directive is amended by Directive 2004/101/EC of the European Parliament and of the Council of 27 October 2004 (establishing a link between the EU emissions trading scheme and the Kyoto protocol's project mechanisms), OJ EU L 338/18, 13 November 2004.
3. European Commission, *Green Paper on greenhouse gas emissions trading*, 8 March 2000, COM/2000/87, p. 24.
4. Directive 2003/87, art. 4.
5. Directive 2003/87, art. 6(1).
6. Directive 2003/87, art. 6(2)(c) and (d), (and arts 14 and 15).
7. Directive 2003/87, art. 6(2)(e). See as well art. 12(3) of the Directive.
8. Directive 2003/87, art. 7.
9. Directive 2003/87, art. 14(2) and 14(3).
10. Answers to Frequently Asked Questions on Commission Decision 2004/156/EC of 29 January 2004 establishing guidelines for the monitoring and reporting of greenhouse gas emissions pursuant to Directive 2003/87/EC (Version: 2 March 2005).
11. According to art. 32, the Directive entered into force on 25 October 2003.
12. Commission Decision of 29 January 2004 establishing guidelines for the monitoring and reporting of greenhouse gas emissions pursuant to Directive 2003/87/EC of the European Parliament and of the Council, Brussels, 29.01.2004, C(2004) 130 final; OJ EU L 59, 26 February 2004.
13. Commission guidelines for monitoring and reporting of greenhouse gas emissions, p. 7.
14. Commission guidelines for monitoring and reporting of greenhouse gas emissions, p. 12.
15. Commission guidelines for monitoring and reporting of greenhouse gas emissions, p. 15: 'Minor sources are those emitting 2.5 ktonnes or less per year or that contribute 5% or less to the total annual emissions of an installation, whichever is the highest in terms of absolute emissions'.
16. Commission guidelines for monitoring and reporting of greenhouse gas emissions, p. 8.
17. Commission guidelines for monitoring and reporting of greenhouse gas emissions, p. 8.
18. Directive 2003/87, art. 14(3).
19. Commission guidelines for monitoring and reporting of greenhouse gas emissions, p. 19.
20. Article 15 of the Directive.
21. Annex V, sub 7, of the Directive.
22. Article 16(2) of the Directive.
23. See for instance: 'Shame, Stigma and Crime: Evaluating the Efficacy of Shaming Sanctions in Criminal Law' (anonymous author), (2003) *Harvard Law Review*, 2186–2207.
24. Whitman, J.Q. (1997–98), 'What is wrong with Inflicting Shame Sanctions?', *The Yale Law Journal*, **107**, pp. 1055–92.
25. It might increase the possibilities for public pressure and even legal action from non-governmental environmental agencies and/or citizens. Tietenberg, T. (2002), 'The Tradable Permits Approach to Protecting the Commons: What have we learned?', in Ostrom, E., Dietz, Th., Dolšák, N., Stern, P.C., Stonich, S., Weber, E.U. (eds), *The Drama of the Commons*, Washington DC: National Academy Press, pp. 197–231.
26. Directive 2003/87, art. 16(3).
27. Peterson, S. (2003), 'Monitoring, Accounting and Enforcement in Emissions Trading Re-

- gimes', *Paper for the OECD Global Forum on Sustainable Development: Concerted Action on Tradable Emissions Permits County Forum*, OECD Paris, 17–18 March 2003, p. 7: the allowance price varied between \$65 and \$220 per tonne SO₂.
28. In June 2005, CO₂ allowances were sold in the Netherlands for €22,50. This is more than half the price of the penalty.
 29. Shavell, S. (1985), 'Criminal Law and the Optimal Use of Non-Monetary Sanctions as a Deterrent', *Columbia Law Review*, 1232–62.
 30. Directive 2003/87, art. 16(3).
 31. Famous scholars like Stewart and Wiener have expressed their doubts on implementing and enforcing the international climate change obligations by the EU; they even stated that international GHG obligations would be more effectively implemented and enforced in the USA than in other countries. Stewart, R.B. and Wiener, J.B. (2003), *Reconstructing Climate Policy*, Washington DC: The AEI Press, p. 86.
 32. On 4 May 2005 the Commission decided on a questionnaire for these reports. Commission decision of 4 May 2005 establishing a questionnaire for reporting on the application of Directive 2003/87/EC, OJ EU L 126/43, 19 May 2005.
 33. Directive 2003/87, art. 30.
 34. Member State states the need, for instance, to report the following question: 'Did the competent authority carry out any independent checks on verified reports? If yes, please describe how additional checks were undertaken and/or how many reports were checked'.
 35. See for example the European CFC Regulation (Regulation 2037/2000/EC), discussed by Peeters, M. (2003), 'Legal feasibility of emissions trading: learning points from emissions trading for ozone-depleting substances', in Faure, M., Gupta, J., Nentjes, A. (eds), *Climate Change and the Kyoto Protocol*, Edward Elgar, pp. 161–2.